Garry Sverlov

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Experience

IntervalZero *(RTOS for industrial automation & medical devices)* intervalzero.com Summers 2025 & 2026

Software Engineering Intern — Real-Time Operating Systems C++, Bash, KVM/QEMU, CMake, WinDbg, Win32 SetupAPI

- **Fixed a customer-reported bug** where upgrading wRTOS (IntervalZero's RTOS for Windows) left the virtual network adapter broken until the machine was rebooted: diagnosed that the old driver stayed loaded in the kernel, then shipped a **C++ MSI custom action** that unloads the stale driver and installs the new one through the Windows SetupAPI.
- Rebuilt and automated the C++ API test suite for wRTOS, using CMake to build the DLLs and drivers needed to cover **APIs that previously couldn't be tested automatically**, **cutting nightly failures from 400+ to ~175**.
- Built a Bash test harness generating **280 VM startup/shutdown permutations** to stress-test vRTOS (IntervalZero's virtualized RTOS), then traced all 28 reproducible failures to a **single concurrency bug**: shared-memory cleanup silently failed under a specific multi-VM shutdown sequence.
- Automated vRTOS environment setup in a **Jenkins pipeline** that builds Ubuntu VMs from cloud images with cloud-init and installs vRTOS on top, turning a 30-minute manual process into a **zero-touch nightly build**.

Daphis Therapeutics *(biotech — longevity research)* daphistx.com Spring 2025

Computer Vision Intern Python, OpenCV, NumPy, SciPy

- Built a **Python computer-vision pipeline** that tracks individual *Daphnia* across video, separating animals that overlap or cross paths using the **MOG2 background subtraction** and **watershed splitting** algorithms, then holding each animal's identity frame to frame with **Kalman-filtered tracking**.
- Selected the **five movement metrics** most predictive of longevity from a *Daphnia* study and computed them as the pipeline's output, giving researchers who dose separate tanks with different vitamins a way to compare each vitamin's effect on longevity.

Projects

App Factory github.com/gsverlov/App-factory

- Took an unreliable open-source **LangGraph + Claude Code** app factory (which turns a prompt into a complete working app) that failed on complex builds and got it shipping reliably, fixing how the orchestrator splits and integrates work across up to 6 parallel dev agents.
- Made the orchestrator generate **task-specific skill files** for each sub-agent at runtime instead of using fixed prompts, and added **code-review and security stages** to the pipeline.

ScholarshipAI *(Anthropic x UofT Hackathon, Honorable Mention)* scholarship-ai--garriksv3.replit.app

- Matched students against **200+ scraped scholarships** by classifying each profile to narrow the pool, running **vector search** to shortlist the top 10, then LLM agent calls to rank the final 5.
- Generated tailored application essays by mining winning essays for strategies tied to each scholarship and student type, feeding those strategies into the **LLM prompts**, then using a **reviewer agent** to pick the strongest of several drafts.

Chrome Extension Fact-Checker github.com/gsverlov/factcheck-extension

- Built a Chrome extension that fact-checks articles inline, extracting page text with Readability.js and using **native messaging** to bridge the browser to a local Python host (since extensions can't run Python directly) that calls the **Claude Agent SDK**, then rendering flagged claims as red and yellow underlines via the CSS Custom Highlight API with hover explanations.

Skills

Languages: C/C++, Python, Java, Bash, PowerShell, SQL

Tools & DevOps: Git, Linux CLI, CMake, MSBuild, CI/CD (Jenkins), KVM/QEMU, cloud-init, WinDbg, Wireshark

Libraries & Frameworks: OpenCV, NumPy, SciPy, LangGraph, LLM agents (Claude API / Agent SDK), vector search, React, Node.js

Education

University of Toronto Toronto, ON Expected Graduation: 2028

Bachelor of Science, Computer Science (Specialist) ***GPA: 3.6/4.0***

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Machine Learning, Software Engineering Principles, Systems Programming in C, Databases (SQL)